

Step 7: Testing and Refinement

Test Cases based on input combinations:

Test Case	Train Sensor	Vehicle Sensor	Exected Output	Actual Output	Notes/Improvements
1	Train approaching	No vehicle	Gates lowered, lights on	Gates lowered, lights on	Works as expected
2	No Train	Vehicle on track	Gates lowered, lights on	Gates lowered, lights on	Works as expected
3	Train approaching	Vehicle on track	Gates lowered, lights on	Gates lowered, lights on	Priority safety confirmed
4	Train Passed	No vehicle	Gates raised, lights off	Gates raised, lights off	Works as expected

Refinements

- Add an emergency manual override for railway staff.
- Include redundant sensors to prevent false negatives.
- Introduce delay before raising gates to ensure extra safety margin.